

APURVA PATEL

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Professional Summary

Full-Stack Machine Learning Engineer — Data Scientist — AI Engineer

Cancer research first drew me into AI; since then I've built and owned end-to-end AI systems at a cybersecurity startup, **2+ Y.O.E**, driving product innovation and achieving **\$1M+ ARR**.

Professional Experience

Dune Security(\$8M seed round)

Mar 2025 – Present

AI Engineer

New York City, NY

- Engineered an **Agentic AI framework** enabling autonomous phishing simulation generation for SAT cybersecurity training, orchestrated via **LangGraph**, serving 150K+ users.
- Developed and fine-tuned a **BERT-based classifier** to replace LLaMA 3.1 8B for internal knowledge checks, enhancing performance and efficiency.
- Developed a knowledge-check pipeline utilizing **AWS Lambda**, **Bedrock**, and **MongoDB**, with logs stored in **S3**, all within a compliant VPC.
- Constructed a Microsoft Entra ingestion pipeline using **Twirp RPC**, implementing auto-scaling to maintain a throughput of **1.08 GB/s**.
- Designed a risk-scoring model leveraging embeddings via **Bedrock + Lambda** to assess user risk based on company, role, and department; results stored in **MongoDB**.

Nokia (Nokia Bell Labs)

Jun 2024 – Aug 2024

Artificial Intelligence/Machine Learning Intern

New Providence, NJ

- Designed and developed an **Autoencoder** for 6G Channel data compression, and integrated **MLOps** practices & enhancing model performance with a **20% improvement** in regeneration over ViTs.
- Led the **end-to-end model development** following **SDLC principles** and used **Kubernetes**, **Docker** for system architecture and maintained **CI/CD** pipeline.
- Contributed to NokiaGPT, leveraging **LLM fine-tuning** to develop a **text summarization model** for 3GPP standards documents **reducing 25+ person hours** for the task.

Indian Space Research Organization (SAC)

Dec 2022 – May 2023

AI Researcher

Ahmedabad, India

- Developed a **FPGA** based high-throughput image data acquisition system, enabling faster and accurate space packet **data verification** at a capacity of 1.7 GB/s via **custom python scripts** in LabVIEW.
- Engineered a **CNN** based image processing architecture, achieving 95.73% accuracy while **reducing computation by 45%**, optimizing data processing workflows in mission-critical applications.

Education

Columbia University

Sept 2023 – Dec 2024

Master of Science in Electrical Engineering (GPA: 3.97/4.00)

New York, NY

- Coursework:** Applied ML(A+), Deep Learning(A+), Big Data Analytics(A), Statistics(A+), Generative AI(A), NLP(A)
- Research:** Research on use of **VAEs** to control autonomous vehicles, also achieved a 32% reduction in data volume through data distillation.
- Teaching:** Conducted classes and tutorials for 100+ students for **Big Data Analytics & Data Visualization**

Vellore Institute of Technology

Jul 2019 – May 2023

Bachelor of Technology in Electronics and Instrumentation Engineering (GPA: 3.97/4.00)

Vellore, India

- Dean's list merit scholarship holder for 4 consecutive years, ranked 4/155 students.

SKILLS

Languages	Python, R, SQL, Golang, MATLAB, C, C++
Frameworks	PyTorch, HuggingFace (transformers, trl), LangGraph, Numpy, Pandas, W&B
Cloud	AWS (Lambda, Bedrock, SageMaker, EC2, ELB, S3, DynamoDB), GCP (Vertex AI)
DevOps	Kubeflow, Airflow, Spark, Kafka, Modal, Kubernetes, Docker, CI/CD, Git, ETL
Databases	PostgreSQL, MongoDB, CosmosDB, ChromaDB, Weaviate, Pinecone

Projects

Medical Chatbot with RAG Architecture | Llama-3, Hugging Face, LangChain, HPC

Dec 2024

- Developed a **RAG pipeline** integrating finetuned LLMs (Llama-3, Llama-2, Gemma 1.1, Mistral-7B, DistilGPT2) with LangChain and leveraging **Weaviate** to index/retrieve data from PubMedQA.
- Achieved a **precision of 88.3%** and **BERT score of 0.87**, with memory efficiency at **4.1GB** for low-cost compute.

Real-Time Credit Card fraud detection system on AWS | Lambda, Sagemaker, Flask, Pandas

May 2024

- Deployed a **real-time fraud detection** pipeline on AWS using **Sagemaker** and **AWS Lambda**, leveraging **SMOTE**, **Random Undersampling & Oversampling** to handle extreme class imbalance.
- Trained and optimized **Decision Trees**, **Random Forest**, and **XGBoost** using stratified cross-validation, achieving a maximum **AUC of 0.9938**, ensuring **high recall** while **reducing false negatives**.